MUHAMAD ARIF

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Informatics graduate from Gunadarma University with hands-on experience in Internet of Things (IoT), Artificial Intelligence, and Computer Vision through internship and academic projects. Completed an undergraduate thesis on a real-time PPE violation detection system integrating YOLOv11m, Python, Streamlit, IP Camera/RTSP, Telegram Bot, and ESP32. Experienced in hardware-software integration, real-time monitoring, device testing, HTTP communication, and technical troubleshooting, with a strong interest in IoT and AI-based system development.

EDUCATION

Gunadarma University - Depok Sept 2022 - Present

Bachelor of Informatics — Expected Graduation 2026

- GPA: 3.56/4.00

INTERNSHIP EXPERIENCE

Digital & Operational Support Intern Oct 2025 – Jan 2026

Bukit Sunset Hill Villa & Resto and Cafe Ngopi di Kebon Awi – Sukabumi, Indonesia

- Developed and integrated an IoT-based fingerprint attendance prototype with a central server for employee attendance recording and monitoring.
- Contributed to web-based reservation, analytics dashboard, frontend implementation, and MySQL database integration.
- Performed hardware-software integration, field testing, deployment, documentation, and troubleshooting of device-server network connectivity.

PROJECT EXPERIENCE

Real-Time PPE Violation Detection System Based on YOLOv11m 2026

Undergraduate Thesis - Gunadarma University

- Developed a real-time PPE violation detection system using YOLOv11m to detect Helmet, Mask, Person, and Vest.
- Trained the model using 4,410 images and achieved 93.0% Precision, 85.9% Recall, 91.8% mAP@0.5, and 65.8% mAP@0.5:0.95 on the test set.
- Implemented real-time monitoring through Streamlit using an IP camera stream via RTSP and ONNX model inference.
- Integrated Telegram Bot notifications and an ESP32-controlled IoT buzzer to provide image-based and physical alerts when PPE violations were detected.

Fingerprint-Based Attendance & Door Access Control System 2026

Freelance IoT Project

- Developed a fingerprint-based IoT system using ESP32 for user authentication, attendance recording, and classroom access control.
- Integrated a fingerprint sensor, relay, and solenoid door lock to automatically unlock the door after successful user verification.
- Integrated attendance records with a MySQL database and performed hardware-software testing and troubleshooting.

SKILLS

Hard Skills : IoT System Integration (ESP32, Fingerprint Sensor, Relay, Solenoid Door Lock, Buzzer/Actuator), Hardware-Software Integration, IoT System Design & Workflow, HTTP/API Communication, AI & Computer Vision (YOLOv11, Ultralytics, OpenCV, ONNX), Python for AI/Computer Vision Implementation, MySQL Database Integration, Streamlit, RTSP/IP Camera, Telegram Bot API, Arduino IDE, Google Colab, VS Code, GitHub

Soft Skills : Analytical Thinking & Problem-Solving, Team Collaboration, Leadership, Adaptability, Time Management